Homework 7

EE 999 — Introduction to Control Systems

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Due Wednesday, November 4, 2026

Problem 1 — Review questions 4.1 and 4.2

4.1 Give three advantages of feedback in control.

Feedback can:

- reduce the steady state error inherent to a system
- reduce the impact of disturbances to a system
- provide stability to unstable systems

4.2 Give two disadvantages of feedback in control.

Feedback typically requires:

- more logical complexity
- physical sensors (cost association)

Problem 2 — Text problem 4.6

Consider the DC motor control system with rate (tachometer) feedback shown in Fig. 4.26(a).

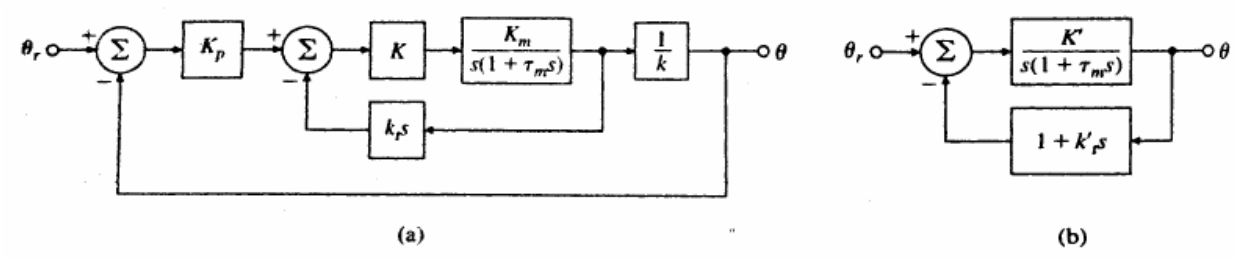


Figure 4.26
Control system for Problem 4.6

Figure 1: Text figure 4.26 (8th Ed.).

Problem 2.A

Find values for K' and k_t' so that the system of Fig. 4.26(b) has the same transfer function as the system of Fig. 4.26(a).

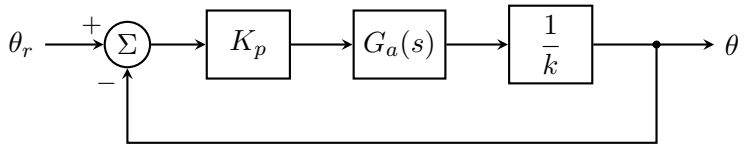


Figure 2: Fig. 4.26(a) with the inner summing junction and tachometer loop combined into the single block $G_a(s)$.

$$G_a(s) = \frac{KK_m}{s(1+\tau_m s)} \frac{1}{1 + k_t s \frac{KK_m}{s(1+\tau_m s)}} \tag{1}$$

$$= \frac{KK_m}{s(1+\tau_m s)} \frac{s(1+\tau_m s)}{s(1+\tau_m s) + k_t s KK_m} \tag{2}$$

$$= \frac{KK_m}{s(1+\tau_m s) + KK_m k_t s} \tag{3}$$

$$\frac{\theta_a}{\theta_r} = \frac{K_p \frac{1}{k} \frac{KK_m}{s(1+\tau_m s)+KK_m k_t s}}{1 + K_p \frac{1}{k} \frac{KK_m}{s(1+\tau_m s)+KK_m k_t s}} \tag{4}$$

$$= \frac{K_p KK_m}{k\left(s(1+\tau_m s) + KK_m k_t s\right) + K_p KK_m} \tag{5}$$

$$= \frac{K_p KK_m}{k\tau_m s^2 + k(1 + KK_m k_t)s + K_p KK_m} \tag{6}$$

$$\frac{\theta_b}{\theta_r} = \frac{\frac{K'}{s(1+\tau_m s)}}{1 + (1 + k_t' s)\frac{K'}{s(1+\tau_m s)}} \tag{7}$$

$$= \frac{K'}{s(1 + \tau_m s) + (1 + k_t' s)K'} \tag{8}$$

$$= \frac{K'}{\tau_m s^2 + (1 + K' k_t')s + K'} \tag{9}$$

$$\frac{K_p K K_m}{k\tau_m s^2 + k(1 + K K_m k_t)s + K_p K K_m} = \frac{K'}{\tau_m s^2 + (1 + K' k_t')s + K'} \tag{10}$$

$$\frac{K_p K K_m / k}{\tau_m s^2 + (1 + K K_m k_t)s + K_p K K_m / k} = \frac{K'}{\tau_m s^2 + (1 + K' k_t')s + K'} \tag{11}$$

$$K' = \frac{K_p K K_m}{k} \tag{12}$$

$$K' k_t' = \frac{K_p K K_m}{k} k_t' = K K_m k_t \tag{13}$$

$$k_t' = \frac{k k_t}{K_p} \tag{14}$$

Problem 2.B

Determine the system type with respect to tracking θ_r and compute the system K_v in terms of parameters K' and k_t'.

$$E_{cl} = \theta_r - \theta_b \tag{15}$$

$$= \left(1 - \frac{\theta_b}{\theta_r}\right)\theta_r \tag{16}$$

$$= \left(1 - \frac{K'}{\tau_m s^2 + (1 + K' k_t')s + K'}\right)\theta_r \tag{17}$$

$$= \left(\frac{\tau_m s^2 + (1 + K' k_t')s + K' - K'}{\tau_m s^2 + (1 + K' k_t')s + K'}\right)\theta_r \tag{18}$$

$$= \left(\frac{s(\tau_m s + 1 + K' k_t')}{\tau_m s^2 + (1 + K' k_t')s + K'}\right)\theta_r \tag{19}$$

Check against a step function:

$$e_{ss} = \lim_{s\to 0} s \, \frac{1}{s} \, \frac{s(\tau_m s + 1 + K' k_t')}{\tau_m s^2 + (1 + K' k_t')s + K'} \tag{20}$$

$$= \lim_{s\to 0} \frac{s(\tau_m s + 1 + K' k_t')}{\tau_m s^2 + (1 + K' k_t')s + K'} \tag{21}$$

$$= \frac{0}{K'} = 0 \tag{22}$$

Check against a ramp function:

$$e_{ss} = \lim_{s\to 0} s\,\frac{1}{s^2}\,\frac{s(\tau_m s + 1 + K'k'_t)}{\tau_m s^2 + (1 + K'k'_t)s + K'} \tag{23}$$

$$= \lim_{s\to 0} \frac{\tau_m s + 1 + K'k'_t}{\tau_m s^2 + (1 + K'k'_t)s + K'} \tag{24}$$

$$= \frac{1 + K'k'_t}{K'} \tag{25}$$

The system is type 1.

$$K_v = \frac{1}{e_{ss}} = \frac{K'}{1 + K'k'_t} \tag{26}$$

Problem 2.C

Does the addition of tachometer feedback with positive k_t increase or decrease K_v?

From equation (12) we have

$$K' = \frac{K_p K K_m}{k} \tag{12}$$

and from equation (14) we have

$$k'_t = \frac{k k_t}{K_p} \tag{14}$$

$$K_v = \frac{K'}{1 + K'k'_t} \tag{27}$$

$$= \frac{\frac{K_p K K_m}{k}}{1 + \frac{K_p K K_m}{k}\frac{k k_t}{K_p}} \tag{28}$$

$$= \frac{\frac{K_p K K_m}{k}}{1 + K K_m k_t} \tag{29}$$

$$= \frac{K_p K K_m}{k(1 + K K_m k_t)} \tag{30}$$

K_v decreases as positive k_t grows in magnitude.

Problem 3 — Working with a control loop and poles and zeros

For a system described by the block diagram of figure 3.

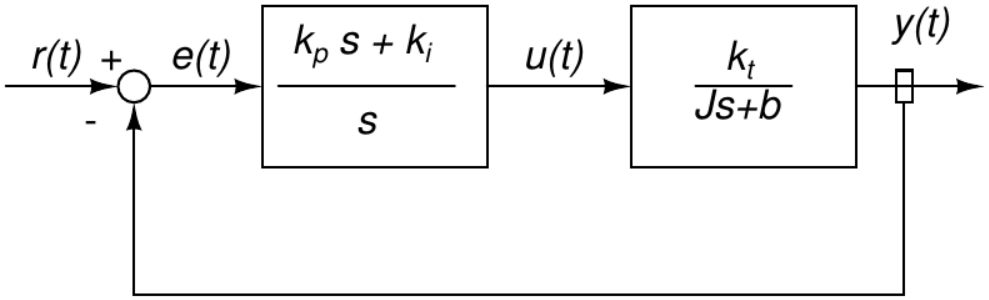


Figure 3: Block diagram of motor with control.

Problem 3.A

Form the transfer function $T_{ry}(s)$.

$$G_1 = \frac{k_p s + k_i}{s} \tag{31}$$

$$G_2 = \frac{k_t}{Js + b} \tag{32}$$

$$Y(s) = \frac{G_1 G_2}{1 + G_1 G_2} R(s) \tag{33}$$

$$= \frac{\frac{k_p s + k_i}{s}\frac{k_t}{Js+b}}{1 + \frac{k_p s + k_i}{s}\frac{k_t}{Js+b}} R(s) \tag{34}$$

$$= \frac{k_t(k_p s + k_i)}{s(Js + b) + k_t(k_p s + k_i)} R(s) \tag{35}$$

$$= \frac{k_t(k_p s + k_i)}{Js^2 + (b + k_t k_p)s + k_t k_i} R(s) \tag{36}$$

$$T_{ry}(s) = \frac{Y(s)}{R(s)} = \frac{k_t(k_p s + k_i)}{Js^2 + (b + k_t k_p)s + k_t k_i} \tag{37}$$

Problem 3.B

With parameters $k_p = 1.5$, $k_i = 0.4$, $k_t = 1.8$, $J = 15$, $b = 2.0$, determine the numerical values for the numerator and denominator of $T_{ry}(s)$.

$$T_{ry}(s) = \frac{(1.8)\big((1.5)s + (0.4)\big)}{(15)s^2 + \big((2.0) + (1.8)(1.5)\big)s + (1.8)(0.4)} \tag{38}$$

$$= \frac{2.7s + 0.72}{15s^2 + (2.0 + 2.7)s + 0.72} \tag{39}$$

$$= \frac{2.7s + 0.72}{15s^2 + 4.7s + 0.72} \tag{40}$$

Problem 3.C

What are the poles and zeros of $T_{ry}(s)$?

Code: `hw_07_3C.m`

```
% Poles and zeros of T_ry(s) = (2.7s + 0.72) / (15s^2 + 4.7s + 0.72)
num = [2.7 0.72];
den = [15 4.7 0.72];

zeros_Try = roots(num)
poles_Try = roots(den)
```

Output: `output/hw_07_3C.txt`

```
zeros_Try = -0.2667
poles_Try =

  -0.1567 + 0.1532i
  -0.1567 - 0.1532i
```

Problem 3.D

Based on the closed-loop poles, determine these four quantities. For each, write the expression that is used to calculate an approximate value from the pole locations.

$$p_{1,2} = -\alpha \pm j\omega \tag{41}$$

$$\omega_n = \sqrt{\alpha^2 + \omega^2} \tag{42}$$

$$\rho = \frac{\alpha}{\omega_n} \tag{43}$$

Code: `hw_07_3D.m`

```
den = [15 4.7 0.72];
p = roots(den);

alpha = -real(p(1))
omega = abs(imag(p(1)))

omega_n = sqrt(alpha^2 + omega^2)
rho = alpha/omega_n

t_r = 1.8/omega_n

M_p = 100*exp(-pi*alpha/omega)

t_s = 4.6/alpha
```

Output: `output/hw_07_3D.txt`

```
alpha = 0.1567
omega = 0.1532
omega_n = 0.2191
rho = 0.7151
t_r = 8.2158
M_p = 4.0208
t_s = 29.362
```

D.1) T_r, the rise time

$$t_r \approx \frac{1.8}{\omega_n} \tag{44}$$

$$\approx 8.22 \text{ s} \tag{45}$$

D.2) ρ, the damping factor

$$\rho = \frac{\alpha}{\omega_n} \approx 0.715 \tag{46}$$

D.3) P.O., the percent overshoot

$$M_p = 100\% \times e^{-\pi \frac{\alpha}{\omega}} = 100\% \times e^{-\frac{\pi\rho}{\sqrt{1-\rho^2}}} \tag{47}$$

D.4) T_s, the settling time

$$t_s \approx \frac{4.6}{\alpha} = \frac{4.6}{\rho\omega_n} \tag{48}$$

$$\approx 29.4 \text{ s} \tag{49}$$

Problem 3.E

In Matlab, setup the system object, plot the step response, and directly measure the four quantities above.

Code: `hw_07_3E.m`

```
if exist('OCTAVE_VERSION', 'builtin')
    pkg load control;
end

num = [2.7 0.72];
den = [15 4.7 0.72];
sys = tf(num, den)

step(sys)
info = stepinfo(sys, 'SettlingTimeThreshold', 0.01)
pole(sys)
zero(sys)
```

Output: `output/hw_07_3E.txt`

```
Transfer function 'sys' from input 'u1' to output ...

          2.7 s + 0.72
 y1:  ---------------------
      15 s^2 + 4.7 s + 0.72

Continuous-time model.
info =

  scalar structure containing the fields:

    RaiseTime = [](0x0)
    RiseTime = 6.6790
    TransientTime = 25.849
    SettlingTime = 25.849
    SettlingMin = 0.9000
    SettlingMax = 1.0748
    Overshoot = 7.4782
    Undershoot = 0
    Peak = 1.0748
    PeakTime = 14
    StaticGain = 1
    InitialJump = 0

ans =

  -0.1567 + 0.1532i
  -0.1567 - 0.1532i

ans = -0.2667
```

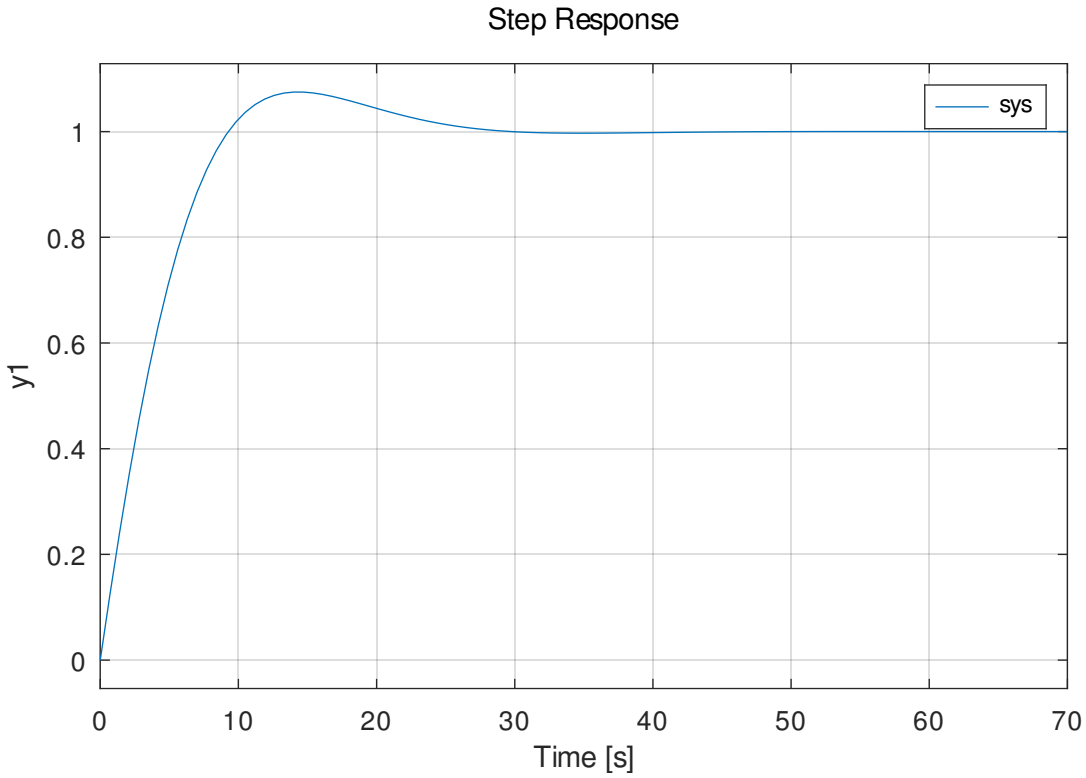


Figure 4: Step response of $T_{ry}(s)$.

Problem 3.F

In parts D and E you computed performance characteristics in two ways, (a) by location of the dominant poles, and (b) by computing and measuring the step response. These two approaches are characterized by (i) one approach is approximate, but provides equations that can be used for design; (ii) one approach is exact, but applies only to that specific system, and generally does not provide design insight. Match each of methods a and b to characterizations i and ii.

(a) is the approximate method (i), which does not take the zeros into account.
(b) includes all of the system information in the plot, so those values are more precise (ii).

Problem 3.G

Now change parameters to $k_p = 5.0$ and $k_i = 1.2$. What are the new closed-loop pole locations? Make a plot of the S-plane. Plot the original closed-loop pole locations (from part C), and the new closed-loop pole locations.

$$T_{ry}(s) = \frac{k_t(k_p s + k_i)}{Js^2 + (b + k_t k_p)s + k_t k_i} \tag{37}$$

$$T_{ry}(s) = \frac{(1.8)\big((5.0)s + (1.2)\big)}{(15)s^2 + \big((2.0) + (1.8)(5.0)\big)s + (1.8)(1.2)} \tag{50}$$

$$= \frac{9.0s + 2.16}{15s^2 + (2.0 + 9.0)s + 2.16} \tag{51}$$

$$= \frac{9.0s + 2.16}{15s^2 + 11s + 2.16} \tag{52}$$

Code: `hw_07_3G.m`

```
if exist('OCTAVE_VERSION', 'builtin')
    pkg load control;
end

num_G = [9 2.16];
den_G = [15 11 2.16];

sys_G = tf(num_G, den_G);

pole_G = pole(sys_G)
zero_G = zero(sys_G)

num_C = [2.7 0.72];
den_C = [15 4.7 0.72];

sys_C = tf(num_C, den_C);

pole_C = pole(sys_C)
zero_C = zero(sys_C)

pzmap(sys_G, sys_C)
legend('Part G poles', 'Part G zeros', 'Part C poles', 'Part C zeros', 'location', 'south')
grid on
```

Output: `output/hw_07_3G.txt`

```
pole_G =

  -0.3667 + 0.0978i
  -0.3667 - 0.0978i

zero_G = -0.2400
pole_C =

  -0.1567 + 0.1532i
```

```
  -0.1567 - 0.1532i

zero_C = -0.2667
```

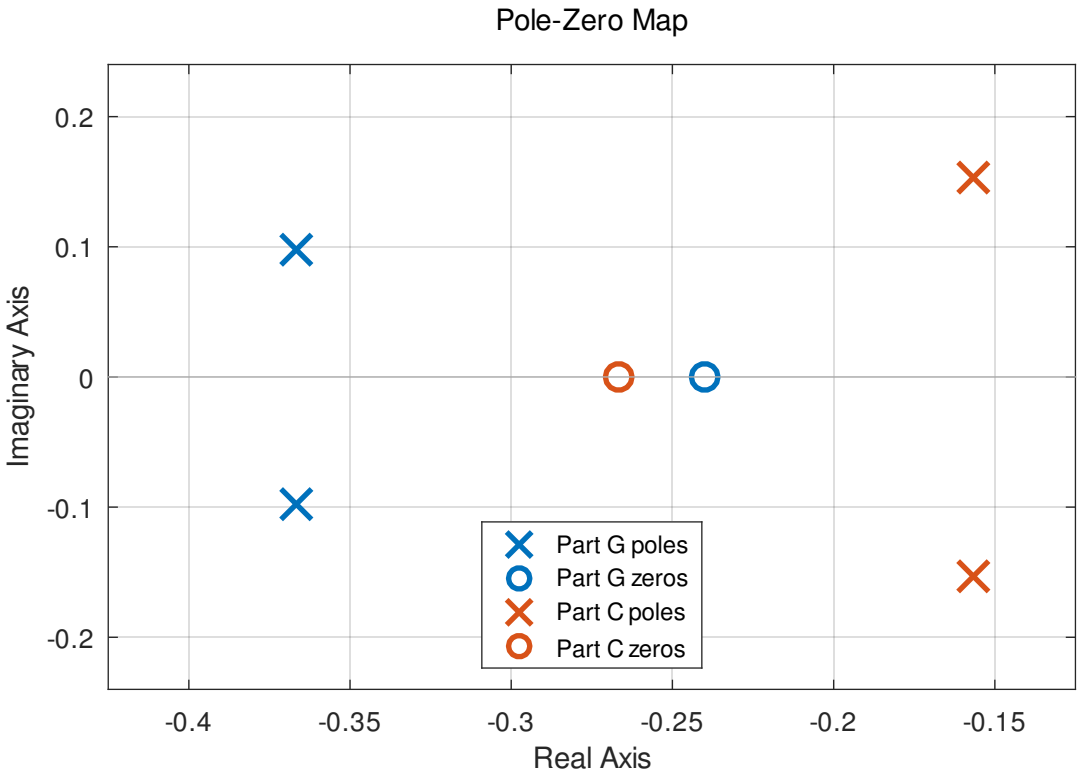


Figure 5: S-plane poles and zeros of $T_{ry}(s)$ for the original (part C) and new (part G) gains.

Problem 3.H

Without calculating any quantities, but looking at how the poles moved, and considering the analysis equations (in part D),

H.1) Do you expect the rise time to be faster or slower?

$$t_r \approx \frac{1.8}{\omega_n} \tag{44}$$

Given ω_n is the distance of the poles from the origin, I would expect t_r to be smaller (faster) with the new values, since the new poles are farther from the origin.

H.2) Do you expect the damping factor to be greater or less?

$$\rho = \frac{\alpha}{\omega_n} \tag{43}$$

I would expect the damping factor to be greater for the Part G equations, since the ratio of the real part to the total length of the vector from the origin to the poles is greater.

H.3) Do you expect the percentage overshoot to be greater or less?

$$M_p = 100\% \times e^{-\pi \frac{\alpha}{\omega}} = 100\% \times e^{-\frac{\pi\rho}{\sqrt{1-\rho^2}}} \tag{47}$$

Since the ratio of the real part to the imaginary part is so much greater with the new values, I would expect the percent overshoot to be less with the new values.

H.4) Do you expect the settling time to be greater or less?

$$t_s \approx \frac{4.6}{\alpha} = \frac{4.6}{\rho\omega_n} \tag{48}$$

I would expect the settling time to be lower, since the magnitude of the real part (α) is greater.

Problem 3.I

In Matlab, setup the new $G_c(s)$ and $T_{ry}(s)$ as in part E, and run `stepinfo(Try)`*. How did the values of T_r, ρ, PO, T_s change relative to what you got in part E? Did the changes agree with your expectations?*

Code: `hw_07_3I.m`

```
if exist('OCTAVE_VERSION', 'builtin')
    pkg load control;
end

num_G = [9 2.16];
den_G = [15 11 2.16];

sys_G = tf(num_G, den_G);

num_C = [2.7 0.72];
den_C = [15 4.7 0.72];

sys_C = tf(num_C, den_C);

step_C = stepinfo(sys_C, 'SettlingTimeThreshold', 0.01)
damp_C = damp(sys_C)

step_G = stepinfo(sys_G, 'SettlingTimeThreshold', 0.01)
damp_G = damp(sys_G)
```

Output: `output/hw_07_3I.txt`

```
step_C =

  scalar structure containing the fields:

    RaiseTime = [](0x0)
    RiseTime = 6.6790
    TransientTime = 25.849
    SettlingTime = 25.849
    SettlingMin = 0.9000
```

```
    SettlingMax = 1.0748
    Overshoot = 7.4782
    Undershoot = 0
    Peak = 1.0748
    PeakTime = 14
    StaticGain = 1
    InitialJump = 0

damp_C =

   0.2191
   0.2191

step_G =

  scalar structure containing the fields:

    RaiseTime = [](0x0)
    RiseTime = 2.7344
    TransientTime = 14.761
    SettlingTime = 14.761
    SettlingMin = 0.9000
    SettlingMax = 1.0566
    Overshoot = 5.6587
    Undershoot = 0
    Peak = 1.0566
    PeakTime = 6.6000
    StaticGain = 1
    InitialJump = 0

damp_G =

   0.3795
   0.3795
```

The new values align with the expectations from part H.

Problem 4 — Text problem 4.7

A block diagram of a control system is shown in Fig. 4.27.

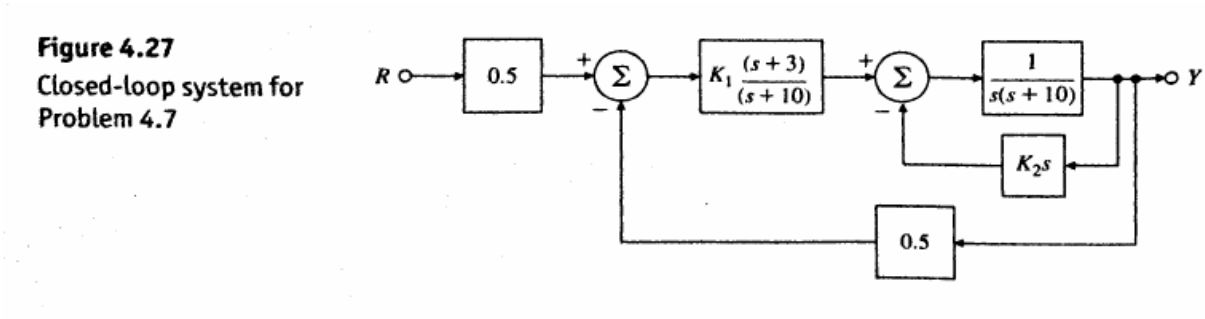


Figure 6: Text figure 4.27 (8th Ed.).

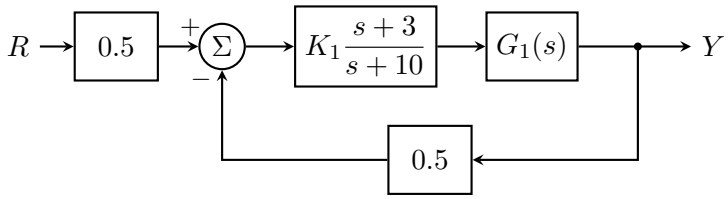


Figure 7: Fig. 4.27 with the inner summing junction and $K_2 s$ feedback loop combined into the single block $G_1(s)$.

$$G_1 = \frac{1}{s(s+10)} \frac{1}{1 + \frac{K_2 s}{s(s+10)}} \tag{53}$$

$$= \frac{1}{s(s+10) + K_2 s} \tag{54}$$

$$= \frac{1}{s(s+10+K_2)} \tag{55}$$

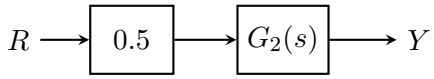


Figure 8: The remaining loop, $K_1 \frac{s+3}{s+10}$ and $G_1(s)$ with the 0.5 feedback, combined into the single block $G_2(s)$.

$$G_2 = \frac{K_1 \frac{s+3}{s+10} G_1}{1 + 0.5 K_1 \frac{s+3}{s+10} G_1} \tag{56}$$

$$= \frac{K_1 (s+3) G_1}{(s+10) + 0.5 K_1 (s+3) G_1} \tag{57}$$

$$= \frac{K_1 (s+3) \frac{1}{s(s+10+K_2)}}{(s+10) + 0.5 K_1 (s+3) \frac{1}{s(s+10+K_2)}} \tag{58}$$

$$= \frac{K_1 (s+3)}{s(s+10)(s+10+K_2) + 0.5 K_1 (s+3)} \tag{59}$$

$$\frac{Y}{R} = 0.5 G_2 \tag{60}$$

$$= \frac{0.5 K_1 (s+3)}{s(s+10)(s+10+K_2) + 0.5 K_1 (s+3)} \tag{61}$$

Problem 4.A

If r is a step function and the system is closed-loop stable, what is the steady-state tracking error?

$$E = R - Y \tag{62}$$

$$= \left(1 - \frac{Y}{R}\right) R \tag{63}$$

$$= \left(1 - \frac{0.5 K_1 (s+3)}{s(s+10)(s+10+K_2) + 0.5 K_1 (s+3)}\right) R \tag{64}$$

$$= \left(\frac{s(s+10)(s+10+K_2) + 0.5 K_1 (s+3)}{s(s+10)(s+10+K_2) + 0.5 K_1 (s+3)} - \frac{0.5 K_1 (s+3)}{s(s+10)(s+10+K_2) + 0.5 K_1 (s+3)}\right) R \tag{65}$$

$$= \left(\frac{s(s+10)(s+10+K_2)}{s(s+10)(s+10+K_2) + 0.5 K_1 (s+3)}\right) R \tag{66}$$

$$e_{ss} = \lim_{s \to 0} s \frac{1}{s} \frac{s(s+10)(s+10+K_2)}{s(s+10)(s+10+K_2) + 0.5 K_1 (s+3)} \tag{67}$$

$$= \lim_{s \to 0} \frac{s(s+10)(s+10+K_2)}{s(s+10)(s+10+K_2) + 0.5 K_1 (s+3)} \tag{68}$$

$$= \frac{0}{1.5 K_1} = 0 \tag{69}$$

Problem 4.B

What is the system type?

$$e_{ss} = \lim_{s\to 0} s\,\frac{1}{s}\,\frac{s(s+10)(s+10+K_2)}{s(s+10)(s+10+K_2)+0.5K_1(s+3)} \tag{67}$$

$$= \lim_{s\to 0} \frac{s(s+10)(s+10+K_2)}{s(s+10)(s+10+K_2)+0.5K_1(s+3)} \tag{68}$$

$$= \frac{0}{1.5K_1} = 0 \tag{69}$$

$$e_{ss} = \lim_{s\to 0} s\,\frac{1}{s^2}\,\frac{s(s+10)(s+10+K_2)}{s(s+10)(s+10+K_2)+0.5K_1(s+3)} \tag{70}$$

$$= \lim_{s\to 0} \frac{(s+10)(s+10+K_2)}{s(s+10)(s+10+K_2)+0.5K_1(s+3)} \tag{71}$$

$$= \frac{10(10+K_2)}{1.5K_1} \tag{72}$$

The system is type 1.

Problem 4.C

What is the steady-state error to a ramp velocity 5.0 if $K_2 = 2$ and K_1 is adjusted to give a system step overshoot of 17%?

write the answer here